3D Reconstruction

Stereo Vision, Camera Calibration, Essential Matrix, and Triangulation

Homework Report

Method	Python, OpenCV, NumPy, Matplotlib
Stereo points	12 corresponding points (P1-P12)
8-point algorithm	P1-P8
3D triangulation	P9-P12

Objective

The objective of this assignment was to reconstruct 3D information from a stereo image pair. The workflow consisted of manually selecting corresponding image points, calibrating the camera with a printed checkerboard, estimating the Essential Matrix using the 8-point algorithm, recovering the relative camera rotation and translation direction, and triangulating a separate set of points into 3D coordinates.

Part A - Stereo Images and Corresponding Points

Two photographs of the same cardboard box were captured from slightly different camera positions while keeping the object stationary. Twelve corresponding physical locations were identified manually in both images. P1-P8 were reserved for the 8-point algorithm, while P9-P12 were kept separate for Part D triangulation.

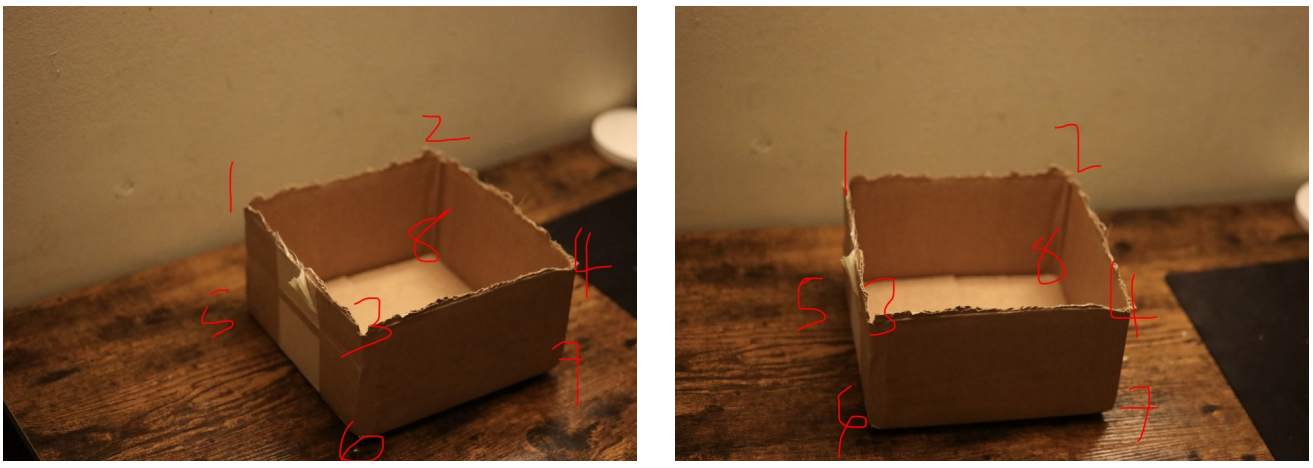


Figure 1. Stereo image pair with manually identified corresponding points.

Point	Image 1 u	Image 1 v	Image 2 u	Image 2 v
P1	557.7	440.2	386.5	396.3
P2	971.4	325.5	872.9	372.5
P3	820.6	723.4	452.0	694.9
P4	1282.4	578.0	1017.6	679.6

Point	Image 1 u	Image 1 v	Image 2 u	Image 2 v
P5	548.9	685.7	377.3	611.1
P6	797.5	970.5	441.2	945.1
P7	1252.3	802.7	980.7	905.0
P8	964.5	551.8	851.4	598.0
P9	666.2	584.9	405.8	539.5
P10	1136.9	442.5	959.9	521.0
P11	1036.1	651.8	727.5	685.7
P12	778.3	390.2	648.2	387.9

Table 1. Manually selected image coordinates for the 12 correspondence points.

Part B - Camera Calibration

A printed checkerboard containing 11 squares across and 8 squares down was used for calibration. OpenCV therefore searched for 10 x 7 = 70 internal checkerboard corners. Seven calibration images were provided in the final set, and five were successfully detected and used by the calibration routine.

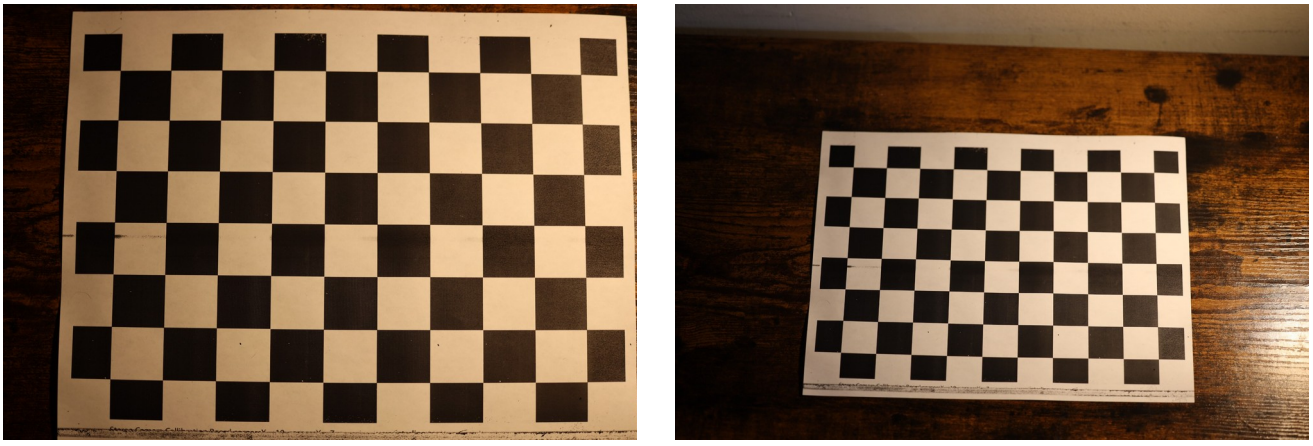


Figure 2. Example checkerboard calibration images from different viewpoints.

Calibration results

Calibration image resolution: 6720 x 4480 pixels

Successful checkerboard detections: 5

Calibration RMS error: 3.3513 pixels

Mean reprojection error: 3.2863 pixels

Original intrinsic matrix K:

[[14502.3918, 0, 3360.0000],
[0, 14505.3738, 2240.0000],
[0, 0, 1]]

Distortion coefficients:

[-0.247286, -5.611287, 0, 0, 58.090498]

Because the stereo images used for Parts C and D were 1536 x 1024 rather than 6720 x 4480, the intrinsic parameters were scaled by 1536/6720 = 1024/4480 = 0.22857.

Scaled intrinsic matrix used for stereo reconstruction:

K = [[3314.8324, 0, 768.0000],
[0, 3315.5140, 512.0000],
[0, 0, 1]]

Part C - Essential Matrix Using the 8-Point Algorithm

The world coordinate system was centered at the first camera pinhole, with the positive Z-axis pointing in the viewing direction. P1-P8 were used to estimate the Essential Matrix.

1. Convert each image point to homogeneous form

For each pixel coordinate p = (u, v), the homogeneous point was written as p_h = [u, v, 1]^T.

2. Normalize the image coordinates

The points were converted to normalized camera coordinates using x = K^-1 p_h.

3. Build the 8 x 9 linear system

For x1 = (x, y, 1) and x2 = (x', y', 1), one row of A was [x'x, x'y, x', y'x, y'y, y', x, y, 1].

4. Solve A e = 0 with SVD

The vector corresponding to the smallest singular value of A was reshaped into the initial 3 x 3 Essential Matrix.

5. Enforce the Essential Matrix singular-value constraint

A valid Essential Matrix has singular values (s, s, 0). The first two singular values were averaged and the third was set to zero.

6. Recover relative camera pose

The corrected E was decomposed into two candidate rotations and two translation signs. Cheirality was used to select the pose with the greatest number of points in front of both cameras.

Example normalized coordinates for P1

Image 1: [-0.063442, -0.021656, 1]^T

Image 2: [-0.115089, -0.034897, 1]^T

The final Essential Matrix was:

E = [[0.169564, 0.672283, -0.010092],
[-0.677118, 0.174246, -0.079696],
[0.054965, 0.112539, 0.001152]]

Final singular values: [0.704169, 0.704169, approximately 0]

Selected rotation matrix:

R = [[0.967575, -0.244967, -0.061562],
[0.246701, 0.968837, 0.022219],
[0.054200, -0.036686, 0.997856]]

Selected translation direction:

t = [-0.174131, 0.036274, 0.984054]^T

The translation vector represents direction only; its magnitude is unknown because two-view reconstruction determines translation only up to an overall scale factor. The epipolar residuals x2^T E x1 were on the order of 10^-3. The cheirality check placed 4 of the 8 estimation points in front of both cameras, indicating that the reconstruction is approximate and sensitive to the manually clicked correspondences and calibration accuracy.

Part D - 3D Triangulation and Point Cloud

P9-P12 were intentionally excluded from the 8-point estimation and were used only for 3D reconstruction. With Camera 1 defined as the world origin, the projection matrices were P1 = [I | 0] and P2 = [R | t].

For each correspondence, linear triangulation formed the system:

A_X = [x1 P1(3,:) - P1(1,:);
y1 P1(3,:) - P1(2,:);
x2 P2(3,:) - P2(1,:);
y2 P2(3,:) - P2(2,:)]

The homogeneous 3D point X was obtained from the right singular vector associated with the smallest singular value of A_X, then divided by its fourth coordinate.

Point	X	Y	Z	Depth check
P9	-7.274579	1.816492	182.577175	front
P10	-2.307201	0.708311	-21.682408	behind
P11	0.582386	0.205707	7.366273	front
P12	0.014812	0.280602	-6.608381	behind

Table 2. Reconstructed 3D coordinates in arbitrary relative scale.

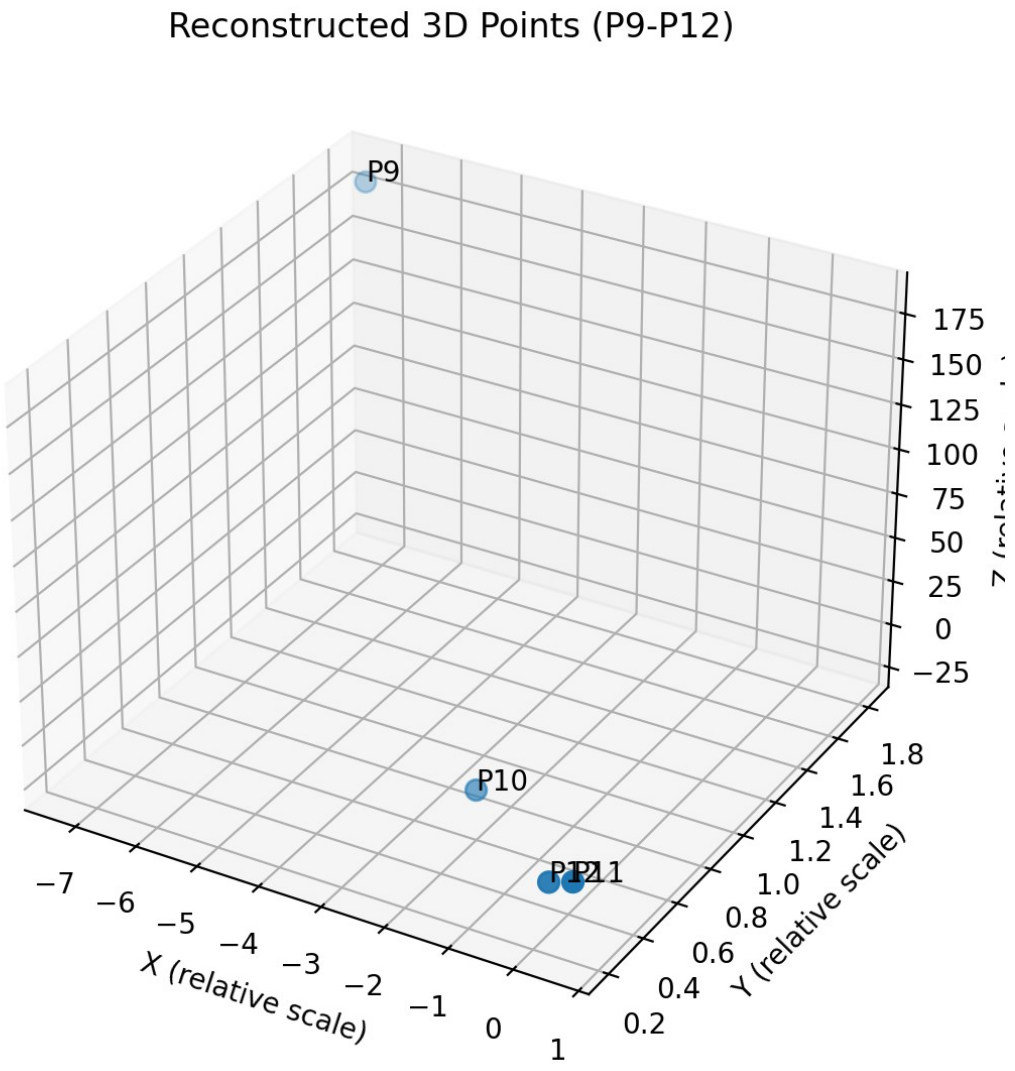


Figure 3. Reconstructed 3D point cloud for P9-P12.

Discussion of discrepancies

The reconstruction did not perfectly match the physical geometry of the box. P9 and P11 had positive depth in both camera coordinate systems, while P10 and P12 produced negative depth. P9 also had a much larger recovered Z value than the other points. These discrepancies are consistent with the assignment warning that reconstruction may fail or differ from the actual scene. Likely sources of error include manual point-selection uncertainty, calibration/reprojection error, residual lens distortion, using exactly eight points for the Essential Matrix, and the sensitivity of linear triangulation to small correspondence errors. In addition, because the translation magnitude is not known, the recovered 3D coordinates are only defined up to scale and should not be interpreted as meters or centimeters.

Conclusion

The complete stereo reconstruction pipeline was implemented successfully: corresponding feature points were identified, the camera was calibrated from checkerboard images, an Essential Matrix

was estimated with the 8-point algorithm, relative rotation and translation direction were recovered, and separate correspondence points were triangulated into a 3D point cloud. The final 3D geometry contained noticeable error, but the results demonstrate the full reconstruction process and illustrate how calibration and correspondence accuracy affect stereo vision.

Appendix - Submitted Code

The implementation is provided in two Python files submitted with the report:
- calibrate.py: checkerboard detection, camera calibration, reprojection error, and intrinsic-matrix scaling.
- main.py: 8-point Essential Matrix estimation, pose recovery, triangulation, depth checking, and 3D point-cloud visualization.